

SEMESTER END EXAMINATIONS – JANUARY 2023

Program	: B.E. – Information Science and Engineering	Semester	: V
Course Name	: Computer Networks	Max. Marks	: 100
Course Code	: IS51	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- A large number of consecutive IP address are available starting at 198.16.0.0/16. Suppose that four organizations, A, B, C, and D, request 4000, 2000, 4000, and 8000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation. CO1 (08)
 - Give reasons for address depletion. Discuss the different classes of classful addressing. CO1 (06)
 - Illustrate the usage of Fragmentation related fields in IPV4 with its datagram packet format. CO1 (06)
- Discuss with diagram, the working of NAT using one IP address. CO1 (06)
 - Describe the different transition strategies from IPV4 to IPV6. CO1 (06)
 - Illustrate the significance of each field present in the IPV6 protocol with its datagram packet format. CO1 (08)

UNIT - II

- Consider the network. Distance vector routing is used, and the following: CO2 (06)
vectors have just come in to router C: from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4). The cost of the links from C to B, D, and E, are 6, 3, and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the cost.

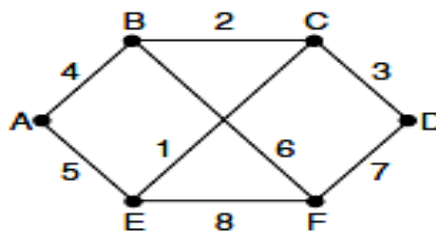


Fig.3(a)

- Explain IGMP message types with their formats. Also discuss IGMP Encapsulation. CO2 (08)
- Find the class of each address: CO2 (06)
 - 00000001 00001011 00001011 11101111
 - 11000001 10000011 00011011 11111111
 - 14.23.120.8
 - 252.5.15.111
 - 01110101 10010101 00011101 00000010
 - 129.11.11.239

4. a) What is DHCP and why is it required? Explain the operation of the same on an internetwork with the help of a schematic diagram. CO2 (06)
- b) For the network given below, explain how link state algorithm is used in calculating the route. Show the shortest path calculation for every edge. The cost from A-B :2, B-C:3, C-F:5, A-D:1, D-E:1, E-F:2, A-C : 5, B-D:2, C-D: 3, C-E:1 CO2 (08)

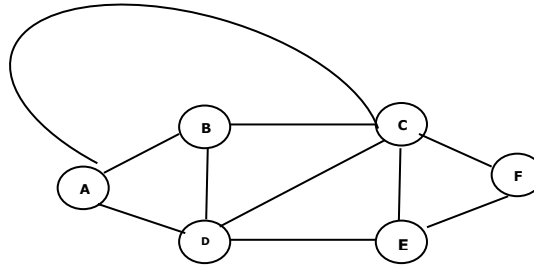


Fig.4(b)

- c) Consider the figure below. How many packets are generated by a broadcast from B, using (i) Reverse path forwarding (ii) The sink tree CO2 (06)

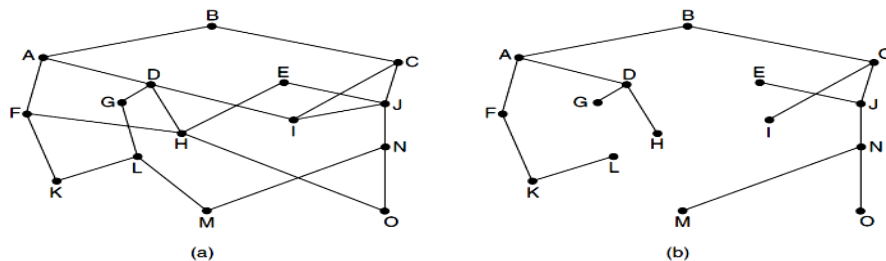


Fig.4(c)

UNIT - III

5. a) Differentiate between the following: CO3 (06)
- i) Connection oriented versus connectionless service.
 - ii) Reliable versus unreliable
 - iii) Multiplexing and demultiplexing.
- b) Summarize the different features offered by the Transmission Control protocol. CO3 (08)
- c) What is congestion in a network? Discuss the policies applied to prevent the congestion before it happens in a network under open loop congestion control. CO3 (06)
6. a) Explain in detail, the process of TCP connection establishment using three-way handshaking. CO3 (06)
- b) Compare between SCTP packet and TCP segment. CO3 (08)
- c) Discuss the two techniques to control the amount and the rate of the traffic sent to the network in traffic shaping mechanism. CO3 (06)

UNIT- IV

7. a) Briefly describe the different functions of network management system. CO4 (08)
- b) Explain the two methods of address resolution in DNS. How does caching increase the efficiency of name resolution? CO4 (07)
- c) Elaborate on the different services offered by the user agent in Electronic Mail. CO4 (05)

8. a) With necessary diagrams, explain the FTP communication over control connection and Data Connection. CO4 (08)
b) Illustrate with a neat diagram, the mechanism of accessing a remote computer using Network Virtual Terminal. CO4 (06)
Discuss the three attributes used by Structure of Management Information (SMI) in handling an object. CO4 (06)

UNIT - V

9. a) Write the algorithm for RSA. Using the same, compute the public key, private key and cipher text for the following data: CO5 (10)
Plain text M=18; Two prime numbers p=5 and q=11; Integer e=7.
b) Discuss the different security services related to message and entity. CO5 (10)
10. a) Illustrate with a neat diagram, the steps of Diffie-Hellman key exchange method. CO5 (08)
b) With necessary diagrams, discuss the following with respect to digital signature: CO5 (12)
(i) Signing the message itself (ii) Signing the digest.
